

Plasma amino acid signature for sarcopenic phenotypes in community-dwelling octogenarians: Results from the Kawasaki Aging Wellbeing Project

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ABSTRACT

Sarcopenia is one of the primary risk factors for various adverse health events in later life. However, its pathophysiology in the very old population remains unclear. Hence, this study aimed to examine whether plasma free amino acids (PFAAs) correlate with major sarcopenic phenotypes (i.e., muscle mass, muscle strength, and physical performance) in community-dwelling adults aged 85–89 years living in Japan. Cross-sectional data from the Kawasaki Aging Well-being Project were used. We included 133 adults aged 85–89 years. In this study, fasting blood was collected to measure 20 plasma PFAAs. Measures for the three major sarcopenic phenotypes included appendicular lean mass assessed by multifrequency bioimpedance, isometric handgrip strength, and gait speed from a 5 m walk at a usual pace. Furthermore, we used phenotype-specific elastic net regression models adjusted for age centered at 85 years, sex, body mass index, education level, smoking status, and drinking habit to identify significant PFAAs for each sarcopenic phenotype. Higher histidine and lower alanine levels were associated with poor gait speed, but no PFAAs correlated with muscle strength or mass. In conclusion, PFAAs such as plasma histidine and alanine are novel blood biomarkers associated with physical performance in community-dwelling adults aged 85 years or older.

1. Introduction

Sarcopenia has been widely inferred as one of the primary risk factors for numerous adverse health events such as mobility disability, falls, hospitalization, poor quality of health, and mortality (Chen et al., 2020; Cruz-Jentoft et al., 2019). Since 2016, sarcopenia has been recognized as an age-related disease (ICD-10-CM M62.84). It is generally diagnosed by the combination of poor muscle mass, muscle strength, and physical functions. Its prevalence rate differs by age, sex, race, medical history, and dwelling settings (e.g., community or nursing home) (Batsis et al., 2015). In community-dwelling older adults, the estimated prevalence rate of sarcopenia increases with age; for example, according to the

European Working Group on Sarcopenia/Asian Working Group on Sarcopenia, it reaches 9.9 % in the youngest old, 15.1 % in the middle old, and 19.4 % in the oldest old (Cruz-Jentoft et al., 2014; Mayhew et al., 2019). Considering that definitions and cutoff points for sarcopenia vary among working groups (Chen et al., 2020; Cruz-Jentoft et al., 2019; Studenski et al., 2014), in pursuit of generalization, rather than the risk of sarcopenia diagnosis, identifying biomarkers for the early detection of decline for each sarcopenia phenotype may be interesting.

Amino acids are the building blocks of proteins, and they fuel various metabolic pathways; thus, their profiling has a potential to identify candidate biomarkers for sarcopenia. A recent meta-analysis demonstrated that the circulation levels of isoleucine, leucine, and tryptophan

Abbreviations: ALM, appendicular lean mass; BCAA, branched chain amino acid; BMI, body mass index; KAWP, Kawasaki Aging Well-being Project; PFAA, plasma free amino acid; SD, standard deviation.

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were lower in individuals with sarcopenia than in those without sarcopenia (Dai et al., 2021). More recently, an epidemiological study with a large sample size showed that high serum levels in the branched chain amino acids, arginine, taurine, and serine were associated with a high incidence of sarcopenia in community-dwelling older adults (Yeung et al., 2022). Additionally, metabolomics technology reveals that among hundreds of metabolites, circulation levels of amino acids such as leucine, isoleucine, and proline are associated with sarcopenia in community-dwelling adults aged 70–80s (Ottestad et al., 2018; Shin et al., 2022). However, with aging, sarcopenia-related traits decline differently; in fact, the rate of change in gait speed cooccurs with the rate of change in muscle strength, but not with that in muscle mass (Kuo et al., 2020; Osawa et al., 2019). The literature shows sarcopenia phenotype-specific amino acids. For example, lower leucine is associated with poor muscle strength in a community-dwelling older women (Yamada et al., 2018). In older adults with functional limitation, branched chain amino acid-related metabolites are associated with muscle mass (Lustgarten et al., 2014). In older adults with frailty and those without, proline and arginine were associated with gait performance (Kameda et al., 2020). Amino acid signatures might differ in sarcopenia-related phenotype. Hence, this study aimed to conduct a hypothesis-free, data-driven discovery-phase investigation to identify plasma free amino acids (PFAAs) that are associated with sarcopenic phenotypes in community-dwelling older adults aged 85 years or older.

2. Materials and methods

2.1. Study population and design

This study used cross-sectional data from the Kawasaki Aging Well-being Project (KAWP) study that was conducted in Kanagawa, Japan. The rationale, design, and data collection have been described elsewhere (Arai et al., 2021). Briefly, 12,906 candidates who were 85–89 years old and free of disability were selected from the basic registration of residents and a long-term care insurance database. Out of 9978 candidates who received an invitation letter, a total of 1026 women and men participated in baseline examination at local hospitals between March 2017 and December 2018. Based on a previous study (Kameda et al., 2020), the sample size was calculated using three different correlations ($r = 0.20, 0.25$, and 0.30), assuming that alpha (type I error) was equal to 0.05 (two-tailed) (Fig. A.1). To detect correlation coefficients >0.25 , 124 analytic samples were required. We selected 133 KAWP participants who had PFAAs, sarcopenia-related measures, and concurrent data on body mass index (BMI), education level, smoking status, and drinking habits (Fig. 1). Compared with those without available data, participants in our study were, on average, less educated and had slightly higher hand grip strength (Table A.1). All participants received an extensive explanation of the study and participated after

providing written informed consent. The study protocol conformed to the principles of the Declaration of Helsinki, with the approval of the Ethics Committee of Keio University School of Medicine (ID: 20160297). The KAWP study was also registered in the University Hospital Medical Information Network Clinical Trial Registry as an observational study (ID: UMIN000026053).

2.2. Measurements

2.2.1. Sarcopenia-related assessment

Muscle strength was measured by hand grip strength in the dominant hand twice (Takei 5401 digital dynamometer; Takei Scientific Instruments Co., Ltd., Tokyo, Japan); the higher value was used for further analysis. Physical performance was measured by a 5-m gait speed at a usual pace in a 9-m distance walk with 2 m acceleration and deceleration periods. Gait speed was assessed twice, and the better result was used for further analysis. Furthermore, using a multifrequency bioelectrical impedance analysis in supine position (InBody S10, InBody Japan, Japan), we assessed muscle mass according to the appendicular lean mass (ALM, the sum of both legs and arms).

2.2.2. Plasma free amino acids

Blood samples were collected in EDTA-2Na tubes (Terumo, Tokyo) in the early morning after fasting for at least 8 h. Blood samples were inverted gently, cooled in ice water, and left for at least 15 min. Plasma samples were collected from the supernatant of blood samples that were centrifuged at 2010 g for 15 min at $4\text{ }^{\circ}\text{C}$. All samples were frozen at $-80\text{ }^{\circ}\text{C}$ within 4-h of collection until measurements. The plasma samples were derivatized with 3-aminopyridyl-N-hydroxysuccinimidyl carbamate (Wako Pure Chemicals, Osaka) and then measured their concentration levels by liquid chromatography-mass spectrometry (LC-MS, Shimadzu UF-Amino station, Shimadzu Corporation, Kyoto, Japan). The details of amino-acid measurement were also described (Yamamoto et al., 2016; Yoshida et al., 2015). All measurements were kindly conducted in Research Institute for Bioscience Products and Fine Chemicals, Ajinomoto Co., Inc.

2.2.3. Other variables

We assessed participants' general characteristics, including education level (high school and higher education or not), smoking status (current smoker or not), and drinking habits (yes/no), through face-to-face interviews. In addition, body weight and height were measured while wearing light clothes and standing upright.

2.2.4. Statistical analysis

Data of demographic characteristics are presented as mean $\pm$ SD for continuous variables and percentages for categorical variables. To compare participants in terms of data availability or sex difference (Table A.1), we tested continuous and categorical variables by using Student's *t*-test and the chi-square test, respectively.

We modeled sarcopenia-related traits using the 20 PFAAs adjusted for age centered at 85 years, sex, BMI, education level, smoking status, and drinking habit by elastic net regularized linear regression (R package, eNetXplorer v. 1.1.3) (Candia and Tsang, 2019). The elastic net regression examines the best fitted model by varying the mixing parameter alpha between 0 (ridge regression) and 1 (lasso regression) by steps of 0.1 (Zou and Hastie, 2005). For each alpha, the best shrinkage penalty parameter lambda is selected by a cross validation procedure optimizing the out-of-bag correlation between observed and predicted sarcopenia-related outcomes. For this cross-validation procedure, the modeling was run 1000 times, setting 80 % of samples at random as the training set and the remaining 20 % as the validation set. The best parameter alpha was chosen for models with the highest average correlation between observed and predicted sarcopenia-related outcomes. To estimate significance of model coefficients, eNetXplorer computes null models, randomly permuting sample labels 250 times for each

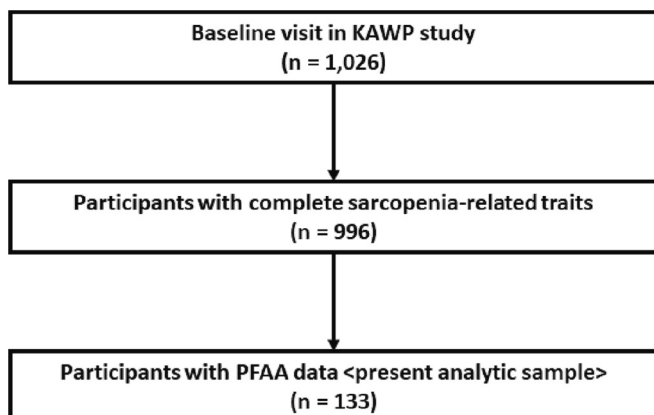


Fig. 1. Flow diagram of participant selection.

cross-validation run. To further explore the association between the two PFAA predictors and gait speed, we followed-up the variable selection process of eNetXplorer with a multiple linear regression model adjusted for the same covariates. A p value of <0.05 was considered statistically significant, and all statistical analyses were performed using R v. 4.2.1.

3. Results

We included 133 participants, of which 45 % were women. Table 1 lists the participant characteristics.

We examined whether a set of 20 PFAAs correlate with each sarcopenic phenotype. In the association between 20 PFAAs and gait speed, the best model performance was obtained for the ridge regression model ($\alpha = 0$). This model used a shrinkage penalty parameter λ (8.78E-05), yielding a Pearson's correlation between the measured and out-of-bag estimated response values equal to a QF of 0.22, which is used as the default quality function (QF) performance metric. Based on the model results on the distribution of null model predictions after random permutations of the response, a p value of 0.03 was considered statistically significant (Fig. 2A). In the model, histidine and alanine were the strongest predictive PFAAs for gait speed ($p < 0.05$; Fig. 2B).

Figs. 3 and 4 present the results of the association between the 20 PFAAs and hand grip strength or ALM. Through α 0 to 1, all model performances significantly fitted these outcomes. Indeed, the empirically estimated p -values saturated at the minimum imposed by the number of random permutations performed, namely $p_{\min} = 1/(1 + 250 \times 1000)$ (Figs. 3A and 4A). However, among the 20 PFAAs, no significant amino acids were found as significantly associated with hand grip strength or ALM ($p > 0.05$; Figs. 3B and 4B).

Based on a follow-up multivariate linear regression model, histidine was found to be strongly associated with gait speed ($p = 0.004$). Meanwhile, there was a weak association between alanine and gait speed ($p = 0.11$; Table 2). Fig. 5 shows the scatter plot between the measured and estimated gait speed in this multiple linear regression model.

Table 1
Participant characteristics.

Characteristic	Overall, n = 133 ^a
Age (years)	87.0 (1.4)
Women, n (%)	60 (45 %)
Education level (high school or higher), n (%)	47 (35 %)
Current drinker, n (%)	53 (40 %)
Current smoker, n (%)	6 (4.5 %)
Body mass index (kg/m ²)	22.9 (3.2)
Hand grip strength (kg)	24.5 (6.5)
Gait speed at a usual pace (m/s)	1.06 (0.23)
Appendicular lean mass (kg)	18.0 (4.6)
Plasma free amino acid (μmol/L)	
Serine	102 (20)
Asparagine	43 (6)
Glycine	206 (44)
Glutamine	589 (67)
Glutamate	36 (14)
Histidine	75 (8)
Threonine	109 (25)
Alanine	330 (79)
Citrulline	42 (12)
Arginine	86 (17)
Proline	143 (45)
Tyrosine	64 (13)
Valine	204 (35)
Methionine	23 (4)
Ornithine	64 (17)
Lysine	179 (26)
Isoleucine	56 (12)
Leucine	106 (20)
Phenylalanine	61 (9)
Tryptophan	48 (9)

^a Mean (SD); n (%).

4. Discussion

This study examined the association between PFAAs and sarcopenic phenotypes in a community-dwelling, very old population. None of the 20 PFAAs correlated with muscle strength or muscle mass. Age, race, and dietary habits of the participants in the present study may be the reasons why these 20 PFAAs could not explain muscle strength or muscle mass. However, our results are interesting in that they demonstrate an association of histidine and alanine with physical performance in the disability-free, very old adults, even after adjusting for possible confounding factors. These results indeed suggest that histidine and alanine may be promising biomarkers of impaired physical performance. A compelling evidence revealed that as a functional aging metric in a hierarchical model of aging, physical performance measures are powerful biomarkers for various health-related outcomes (Ferrucci et al., 2018). For example, slower walking speed is associated with a higher mortality risk (Studenski, 2019; Studenski et al., 2011). Despite the clinical importance of physical performance measures, identifying biomarkers is important to early detect physical decline before reaching a level wherein compensatory mechanisms cease to work (Ferrucci et al., 2020). Although these results need to be replicated in other prospective cohort studies, they shed light on specific biological mechanisms of declining physical performance.

Histidine, the least abundant of amino acids, is involved in tissue growth and repair, and is necessary in the myelin sheath that protects nerve cells (Holecek, 2020). Histidine supplementation improves physical performance, especially high-intensity exercises. This improvement may be explained by the increased intracellular carnosine, a dipeptide composed of two amino acids (β -alanine and histidine), and increased histamine synthesis, which reduces fatigue (Holecek, 2020). Circulation levels of histidine decrease with age. In addition, the blood levels of histidine in the 10th percentile of the current sample were lower than the 2.5th percentile of the reference value for 1890 Japanese adults (Yamamoto et al., 2016). Lower histidine levels are associated with increased inflammatory status, oxidative stress, and increased risk of death (Cheng et al., 2015; Watanabe et al., 2008; Yu et al., 2012). A meta-analysis showed that histidine levels are associated with increased fatigue risk in older adults (Cheng et al., 2015; Watanabe et al., 2008; Yu et al., 2012). In the current study, plasma histidine were associated with physical function in the very old participants. Although the list of significant amino acids for specific outcomes varies by age, race, or dietary habits, the results suggest that histidine may be a key amino acid for reduced exercise capacity in older adults aged over 85 years.

Alanine plays a role in energy metabolism as a substrate for gluconeogenesis via glucose-alanine cycles. However, vast evidence shows the association of alanine aminotransferase, which is an enzyme produced in liver cells, with health-related outcomes including the risk of sarcopenia, frailty, and all-cause mortality (Vespasiani-Gentilucci et al., 2018), the association of alanine with health-related outcomes has been less examined. A recent metabolomics study using 191 plasma metabolites indicated that plasma alanine ranked among the 10 most significant metabolites to distinguish elderly individuals (mean age: 81.9 years) with severe sarcopenia from controls without sarcopenia by matching age and BMI (Shin et al., 2022). Age-related changes in alanine levels differ according to organs. That is, the alanine levels in the muscle and plasma are low (Houtkooper et al., 2011; Uchitomi et al., 2019). Meanwhile, the alanine levels in the white adipose tissues are high (Zhuang et al., 2021). These age-related changes are partly attributed to reduced dietary protein intake, less efficient gluconeogenesis, and physical inactivity (Fukai et al., 2016). While physical performance declines with age, the underlying mechanism for the age-related change in alanine levels has not been fully elucidated yet. With a higher expression of redox-related metabolism occurring in the liver and heart, and lower in the skeletal muscle and brain, age-related, organ-specific metabolic remodeling may be responsible for the dynamic increase in the energy cost of physical activity, resulting in reduced walking speed

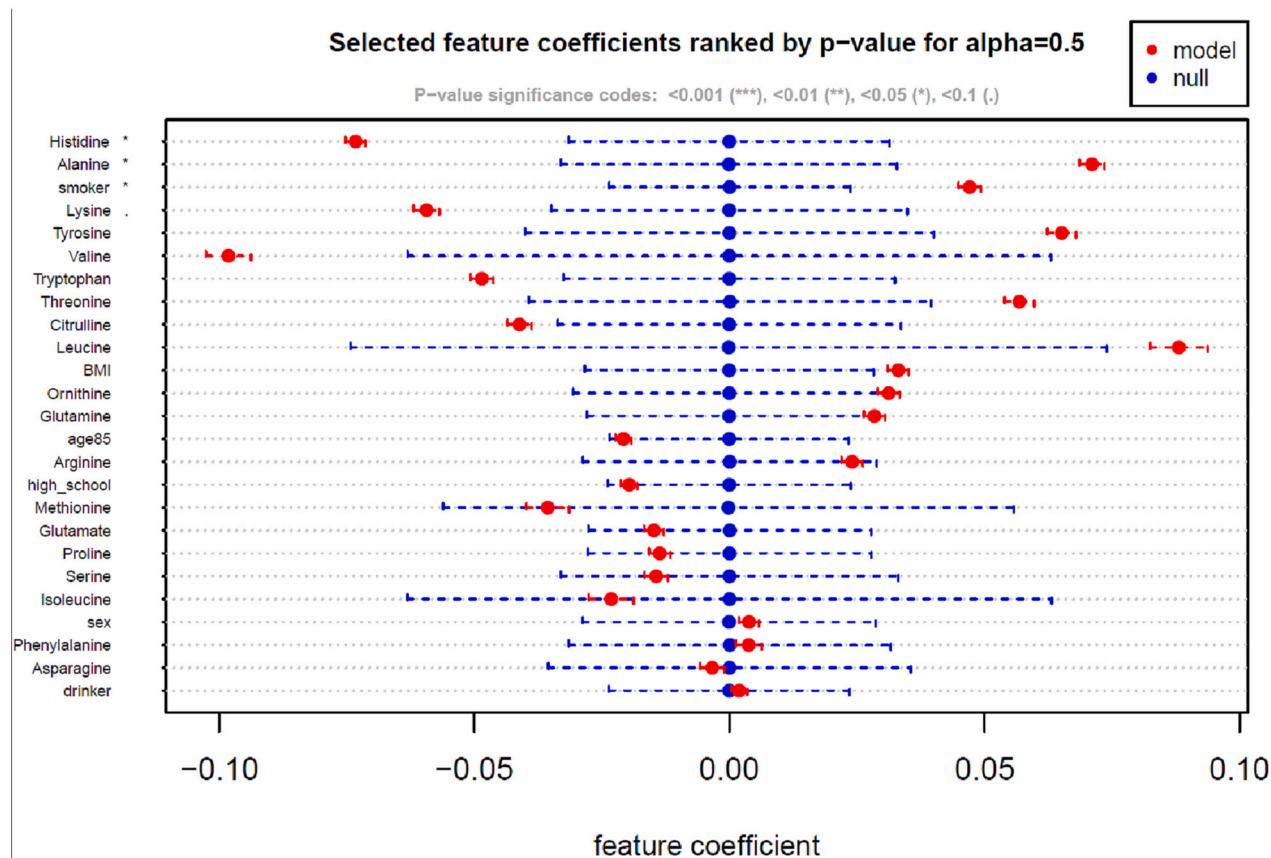
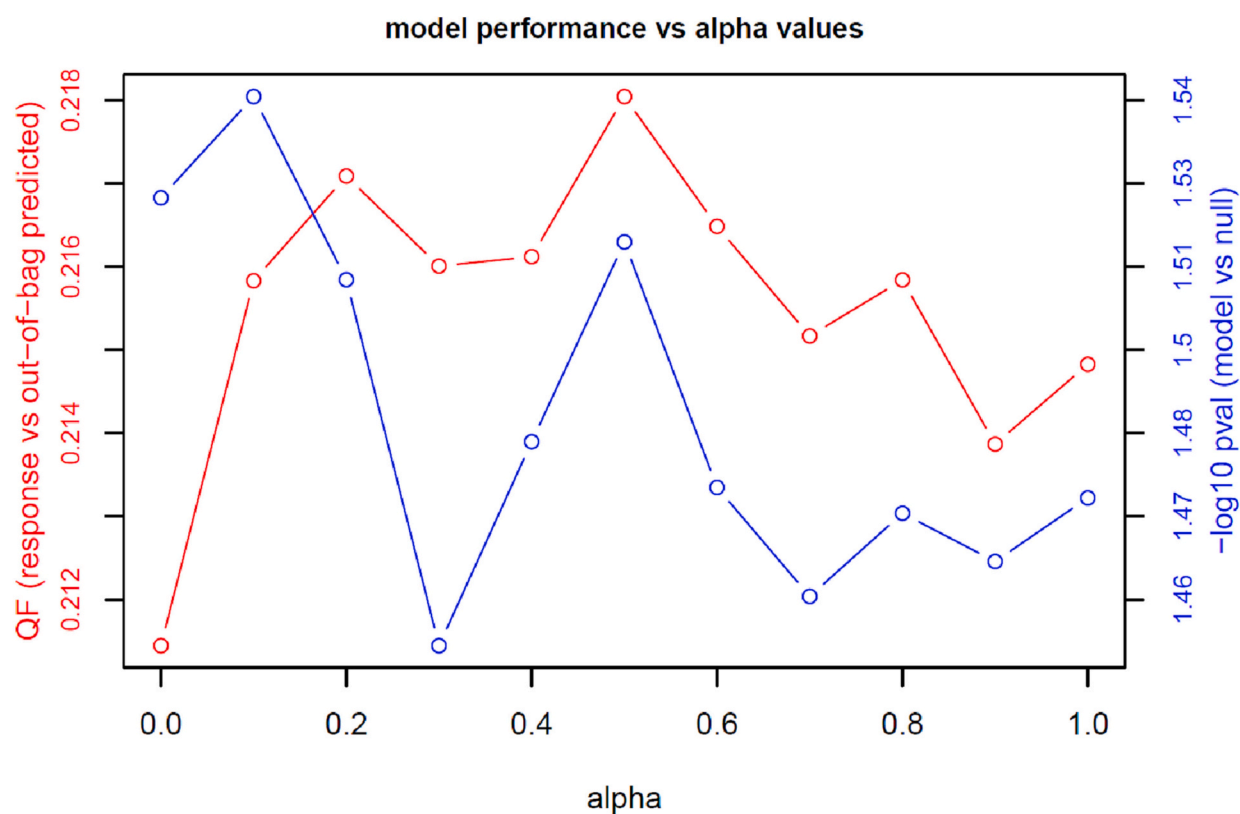


Fig. 2. Linear regression model of the correlation between plasma free amino acids and gait speed.

Upper panel, Performance across α : quality function (QF) and statistical significance (p-value of model vs null). The default QF for linear regression is Pearson's correlation between out-of-bag predictions and the response. Lower panel, Caterpillar plot of feature frequencies. Red symbols (bars) indicate the mean (standard deviation) of model feature frequencies across runs. Blue symbols (bars) denote the mean (standard deviation) of null model feature frequencies across run/permutation null model combinations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

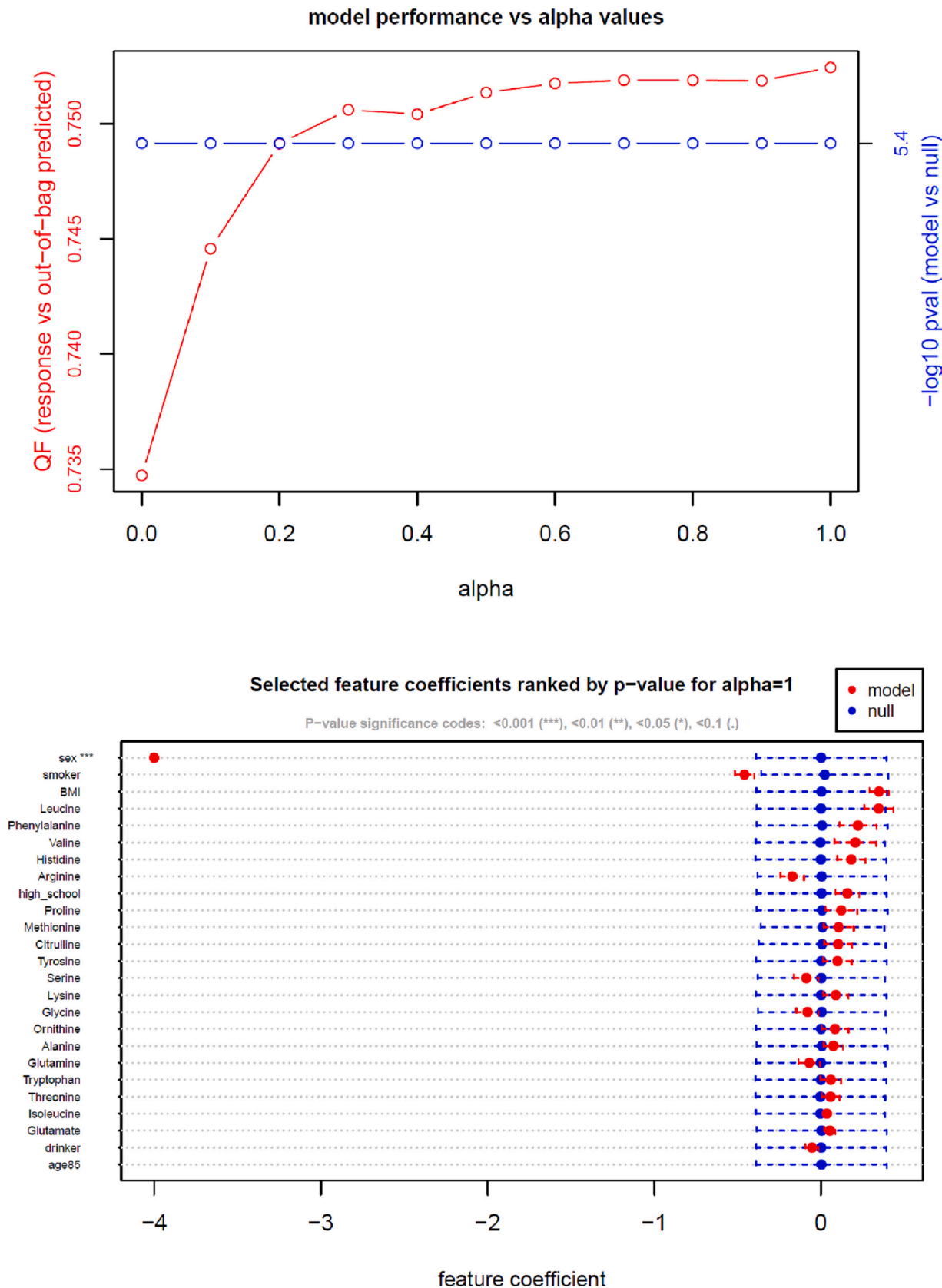


Fig. 3. Linear regression model of the correlation between plasma free amino acids and hand grip strength. Upper panel, Performance across α : quality function (QF) and statistical significance (p-value of model vs null). The default QF for linear regression is Pearson's correlation between out-of-bag predictions and the response. Lower panel, Caterpillar plot of feature frequencies. Red symbols (bars) indicate the mean (standard deviation) of model feature frequencies across runs. Blue symbols (bars) denote the mean (standard deviation) of null model feature frequencies across run/permutation null model combinations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

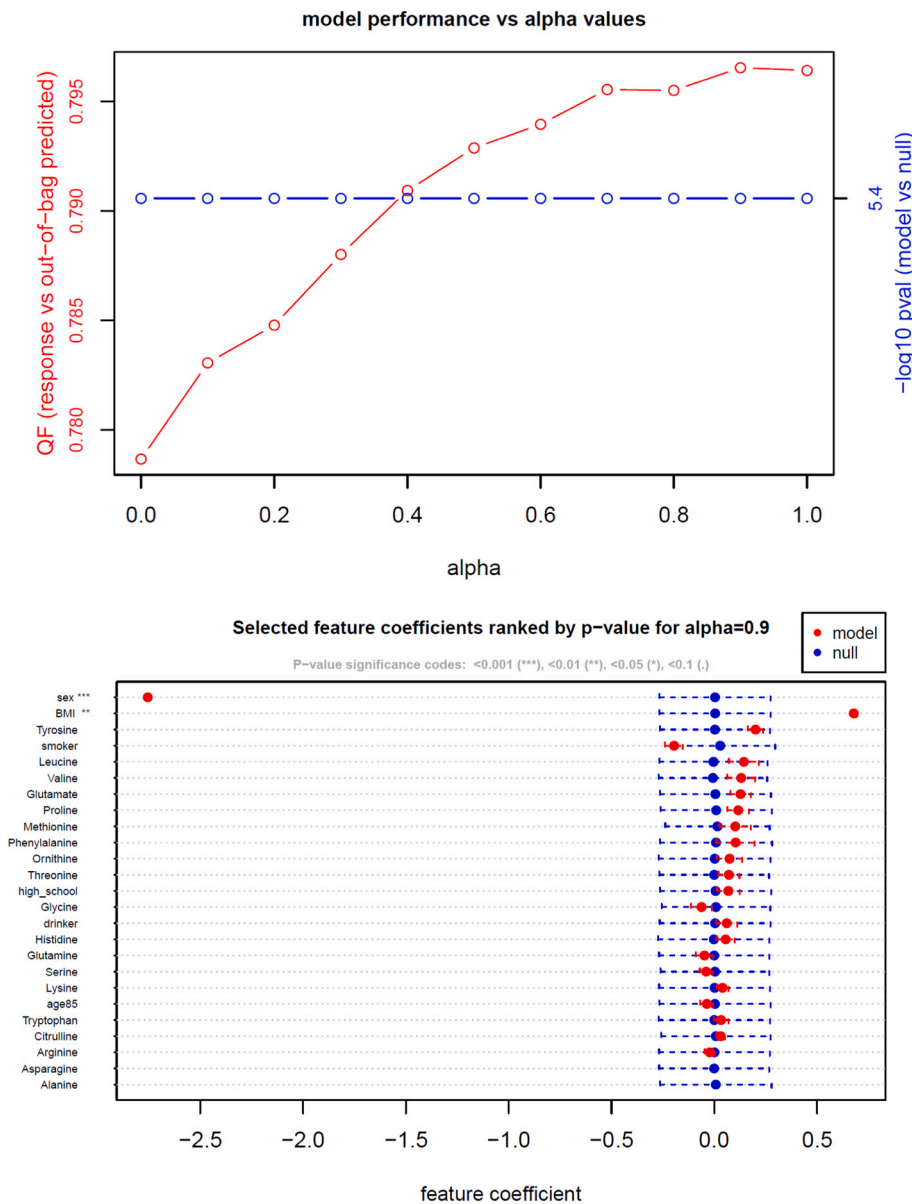


Fig. 4. Linear regression model of the correlation of plasma free amino acids and appendicular lean mass. Upper panel, Performance across α : quality function (QF) and statistical significance (p-value of model vs null). The default QF for linear regression is Pearson's correlation between out-of-bag predictions and the response. Lower panel, Caterpillar plot of feature frequencies. Red symbols (bars) indicate the mean (standard deviation) of model feature frequencies across runs. Blue symbols (bars) denote the mean (standard deviation) of null model feature frequencies across run/permutation null model combinations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Results from a linear regression model of gait speed with two significant amino acids and covariates.

	β	SE	p-value
(Intercept)	1.43	0.25	8.19e-08
Histidine	-0.01	0.003	0.004
Alanine	0.0005	0.0003	0.11
Age	-0.02	0.01	0.28
Sex	0.03	0.05	0.53
Education level	-0.07	0.04	0.11
Drinker	-0.02	0.04	0.56
Smoker	0.19	0.10	0.047
BMI	0.01	0.01	0.38

F-statistic: 2.69; $p = 0.01$; Adjusted R-squared, 0.09.

BMI, body mass index.

(Petr et al., 2021). According to the current study, the correlation coefficient between the measured and estimated gait speed from histidine, alanine and confounders was modest ($r = 0.38$). Results showed that these two amino acids can be a candidate biomarker for detecting poor

mobility. However, physical activity, such as walking, is a complex motor task involving hundreds of interlocking muscles, bones, and joints, as well as the brain and cardiovascular system. Hence, the results also suggest that these amino acids can only capture a piece of walking ability. Additional studies are needed to determine how age-related changes in the circulation levels of alanine may contribute to the decline of such complex motor tasks.

An important strength of our study is that we considered the possible confounders to identify PFAA signatures for sarcopenia phenotypes in octogenarians with similar demographic characteristics. However, our study has some limitations. First, considering that the KAWP participants already underwent a comprehensive health check including physical performance tests, the response rate from the sampling was low (12.6 %), and they were healthy, very old participants. Second, our participants' skeletal muscle index (ALM divided by height squared) was 10 %, which was below the cutoff level in the Asian sarcopenia working group; however, the algorithm of lean mass calculation has not been fully validated among individuals 80 years and older. Further studies are needed to replicate the present results in other cohorts consisting of very old adults by using dual-energy X-ray absorptiometry for ALM

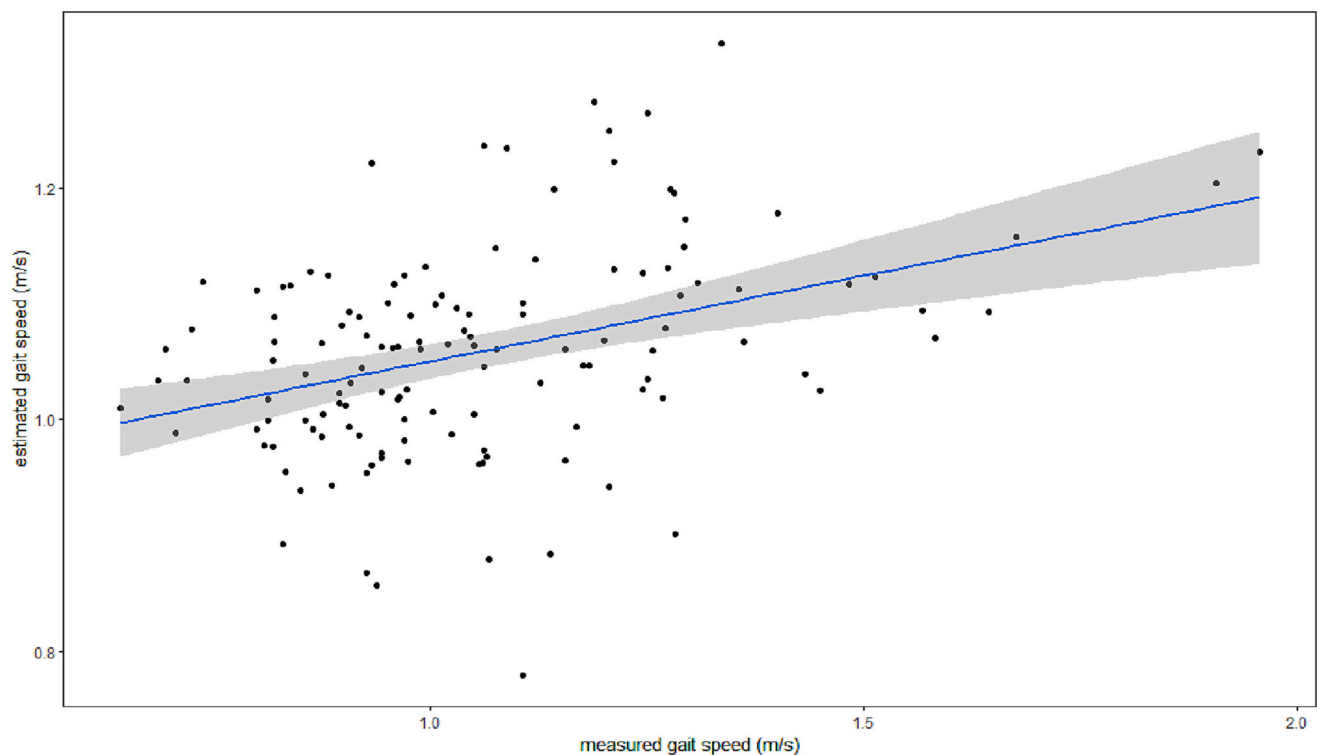


Fig. 5. Correlation between the measured and estimated gait speed in the linear regression model. This graph shows the correlation between the estimated gait speed on the y-axis and the measured gait speed on the x-axis. The Pearson's correlation coefficient was found to be statistically significant ($r = 0.38$, $p = 4.869\text{e-}06$).

assessment. Third, the analysis results did not provide a temporal sequence between PFAAs and sarcopenia-related traits because both measures were obtained at the same time point. Future studies with large sample sizes are needed to examine the temporal sequence of PFAAs and sarcopenia-related traits in a longitudinal study across different health conditions. Finally, the analytic sample size was relatively small compared with that in previous studies (Lustgarten et al., 2014; Yamada et al., 2018; Yeung et al., 2022). Therefore, further studies with larger, independent cohorts of community-dwelling octogenarians in a wider age span should be conducted to validate our findings. However, despite the relatively small cohort size, the elastic net approach presented here is statistically robust. To prevent random chance results, eNetXplorer can be used to implement the hierarchical cross-validation algorithm (as shown in Fig. 2 in the study of Candia and Tsang). First, at the model level, the cross-validated model was compared against an ensemble of cross-validated null models generated from random permutations of the response. The model performance is quantified by a quality function (Pearson's correlation in our case). The comparison between the model's performance and the null model's performance distribution is quantified by an empirical p-value. These results are reported as a function of alpha (Figs. 2(A), 3(A), and 4(A)). Second, at the feature level, a similar procedure is used to quantify the performance of each feature in the model against that feature's performance across the null model ensemble. To identify feature-level performance, the distribution of model regression coefficients is compared against similar distributions from the randomized null model ensemble (Figs. 2(B), 3(B), and 4(B)). This phenotype permutation approach preserves the covariance structure across feature space.

5. Conclusion

In conclusion, this cross-sectional study examined the association of PFAAs with sarcopenic phenotypes in community-dwelling octogenarians. Higher circulation levels of histidine and lower circulation levels of alanine were associated with poor gait speed. Therefore, these amino

acids may be promising biomarkers to assess physical performance in this population.

Software resources

R package eNetXplorer (Candia and Tsang, 2019) v.1.1.3 is available under GNU General Public License GPL-3.0 at <https://github.com/juliencandia/eNetXplorer>.

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CRediT authorship contribution statement

Osawa Y and Arai Y: Study design. **Osawa Y, Arai Y, Abe Y, Oguma Y and Tajima T:** Data collection. **Osawa Y, Candia J, and Arai Y:** Data analysis and interpretation. **Arai Y, Candia J:** Critical revision of the draft. All authors approved the final version of the report.

Declaration of competing interest

None.

Acknowledgement

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.exger.2023.112230>.

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